

Economics and Computation (2026 Fall)

Quiz 1

1. Consider the following two-player payoff-maximization game. Player 1 chooses a row and Player 2 chooses a column.

	Player 2		
	x	y	z
Player 1			
a	(1, 2)	(2, 2)	(5, 1)
b	(4, 1)	(3, 5)	(3, 3)
c	(5, 2)	(4, 4)	(7, 0)
d	(2, 3)	(0, 4)	(3, 0)

- (a) **(20%)** When Player 1 plays c , what is Player 2's best response? Briefly justify your answer.
- (b) **(20%)** Find all pure-strategy Nash equilibria. Explain why each reported outcome is a mutual best response.
- (c) **(20%)** Find all Pareto-optimal outcomes. For every omitted outcome, identify an outcome that Pareto dominates it.
2. Five voters have the following strict preferences over the four outcomes A, B, C , and D :

Voter 1: $B \succ C \succ A \succ D$,

Voter 2: $B \succ D \succ C \succ A$,

Voter 3: $D \succ C \succ A \succ B$,

Voter 4: $A \succ B \succ C \succ D$,

Voter 5: $A \succ C \succ B \succ D$.

Answer both questions below and show the relevant score calculations or pairwise majority comparisons.

- (a) **(20%)** Under the Borda count rule, assign 3, 2, 1, and 0 points to a voter's first-, second-, third-, and fourth-ranked outcomes, respectively. Calculate the total score of every outcome and identify the Borda winner.
- (b) **(20%)** Determine whether a Condorcet winner exists. Justify your conclusion using all pairwise majority comparisons needed to establish the answer.